



8792 - Unlocking the massive stars behind the most spectacular fireworks displays in the early Universe

Cycle: 4, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	Plan 161 all	NIRSpec MultiObject Spectroscopy	(1) rxcj2248_mainphot

ABSTRACT

Many of the most persistent mysteries about the early Universe are wrapped-up in the debate about how globular cluster abundance patterns were imprinted with the signatures of massive star ejecta. The discovery of a population of galaxies powering extremely prominent emission in highly-ionized, nitrogen-enriched gas at $z > 6$ promises to open an entirely new window onto this puzzle and onto the detailed physics of densely-clustered star formation at Cosmic Dawn; yet conclusions thus far are severely limited by the quality of data available for these objects. Here we propose to construct an extraordinarily deep view onto the FUV spectrum of an exemplary nitrogen-rich burst caught in the reionization era, fortuitously

quintuply-lensed by the Abell-S1063 cluster. With a 65 hour stare in G140H, we will capture the first high-fidelity snapshot of the massive stars underlying this emission, testing models for this enrichment by providing constraints on stellar abundances and the presence of very/super-massive stars or stellar populations heavily modified by dense cluster interactions. Simultaneously, we will leverage the rich photometric data for this cluster to fill the MSA with new fainter high-redshift targets, for which we will provide stringent constraints on UV nebular emission and the incidence of these peculiar features to much fainter magnitudes. This lensed galaxy bright enough to be accessible to nebular detections from the ground now represents a singular opportunity for JWST to directly constrain the highly-uncertain primordial massive star populations that underly and confound attempts to model faint nebular emission out to the frontier of Cosmic Dawn.

OBSERVING DESCRIPTION

Our program is built around deep spectroscopy of a quintuply-lensed $z=6.1$ galaxy RXCJ2248-ID. Our science there is dependent upon achieving very high SNR for this galaxy in the restframe far-ultraviolet continuum. To achieve this, we request 65.7 hours science exposure, corresponding to a total charged time of 81.7 hours.

The G140H grating uniquely satisfies our science wavelength (1240-1760Å rest-frame) and spectral resolution ($R>2000$) requirements. The MSA maximizes our signal-to-noise and allows for multiplexing three (and potentially all four) of the images of the primary target simultaneously. The total requested exposure time is based upon ETC calculations for a model spectrum rescaled to match the flux-calibrated lower-resolution JWST spectrum of the median-brightest image. Our simulations suggest that recovering stellar wind feature at high-confidence for an extremely metal-poor stellar population without additional sources will require $\text{SNR}>10$ in the full stack at 1240Å rest, corresponding to >6 in the median image; and a resulting total integration time of 65 hours.

Our goal is to maximize on-source exposure and SNR while minimizing overheads. The ETC indicates that the observations are not background limited or in danger of saturation. Data volume considerations from the APT necessitate that we utilize NRSIRS2 as opposed to NRSIRS2RAPID. Since the primary target clump is effectively unresolved, we plan 3-point nod-in-slitlet observations with a standard 3 shutter slitlet to maximize SNR (the detector gaps will naturally be filled by the inherent offsets between the multiple targeted images). We utilize the maximum-recommended 20 Groups/Int (corresponding to ~ 1500 s) to maximize SNR, and set Integrations/Exp to 6 so that the total exposure duration is just under 10000s. A total of 9 of these exposure sets spread over 3 visits then yields the requested total science integration time of 65.7 hours, with total charged time of 81.7 hours.

Proposal 8792 - Targets - Unlocking the massive stars behind the most spectacular fireworks displays in the early Universe

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	rxcj2248_mainphot	RA: 22 48 43.3584 (342.1806600d) Dec: -44 32 3.95 (-44.53443d) Equinox: J2000		
	Comments: Description=[]				

Proposal 8792 - Observation 1 - Unlocking the massive stars behind the most spectacular fireworks displays in the early Universe

Observation	Proposal 8792, Observation 1: Plan 161 all										Tue Mar 11 23:02:05 GMT 2025
	Diagnostic Status: Warning										
	Observing Template: NIRSpec MultiObject Spectroscopy										
	Comments: The special requirement on APA is not strict but preferred; please see text.										
Diagnostics	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:2) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
	(Visit 1:3) Warning (Form): Overheads are provisional until the Visit Planner has been run.										
Fixed Targets	#	Name		Target Coordinates		Targ. Coord. Corrections			Miscellaneous		
	(1)	rxcj2248_mainphot		RA: 22 48 43.3584 (342.1806600d)							
				Dec: -44 32 3.95 (-44.53443d)							
				Equinox: J2000							
	Comments: Description=[]										
Acquisition	#	Reference Star Bin	Target	Filter	MSA Configuration	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	ETC Wkbk.Calc ID
	1		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
	2		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
	3		SAME	F140X	Auto Acq MSA Config	NRSRAPID	3	1	4	171.788	
Template	TA Method	HFF Readout Mode		Obtain Confirmation Images	Science Aperture	Primary Candidate List	Filler Candidate List		Spectral Overlap Map		Spectral Overlap Threshold
	MSATA	false		No	MSA Center	RXCJ2248-ID Images (4 sources)	allothers (239 sources)		jwst-nirspec-g140h		1.5
Reference Stars											

Proposal 8792 - Observation 1 - Unlocking the massive stars behind the most spectacular fireworks displays in the early Universe

	#	Exposure Specification	MSA Configuration	Nod Pattern	Pointing	Aperture PA	Dispersion Offset (Shutters)	Cross-Dispersion Offset (Shutters)	Total Dithers	Total Integrations	Total Exposure Time
Spectral Elements	1	1 (G140H/F070LP)	c1	3 Shutter Slitlet	342.200074375 Degrees - 44.521049722222 244 Degrees	160.98640270179 152			3	18	26522.602
	2	1 (G140H/F070LP)	c1	3 Shutter Slitlet	342.200074375 Degrees - 44.521049722222 244 Degrees	160.98640270179 152			3	18	26522.602
	3	1 (G140H/F070LP)	c1	3 Shutter Slitlet	342.200074375 Degrees - 44.521049722222 244 Degrees	160.98640270179 152			3	18	26522.602
	4	1 (G140H/F070LP)	c1	3 Shutter Slitlet	342.200074375 Degrees - 44.521049722222 244 Degrees	160.98640270179 152			3	18	26522.602
	5	1 (G140H/F070LP)	c1	3 Shutter Slitlet	342.200074375 Degrees - 44.521049722222 244 Degrees	160.98640270179 152			3	18	26522.602
	6	1 (G140H/F070LP)	c1	3 Shutter Slitlet	342.200074375 Degrees - 44.521049722222 244 Degrees	160.98640270179 152			3	18	26522.602
	7	1 (G140H/F070LP)	c1	3 Shutter Slitlet	342.200074375 Degrees - 44.521049722222 244 Degrees	160.98640270179 152			3	18	26522.602
	8	1 (G140H/F070LP)	c1	3 Shutter Slitlet	342.200074375 Degrees - 44.521049722222 244 Degrees	160.98640270179 152			3	18	26522.602
	9	1 (G140H/F070LP)	c1	3 Shutter Slitlet	342.200074375 Degrees - 44.521049722222 244 Degrees	160.98640270179 152			3	18	26522.602
Special Requirements	Group Visits within 53.0 Days Aperture PA Range 160 to 163 Degrees (V3 21.4254303 to 24.4254303) Visits Same PA MSA Planned Aperture PA 161.0000 to 161.0000 Degrees (V3 22.4254303 to 22.4254303)										